# Robust PI Control With Improved Performance for Nonlinear Systems Under Saturated Actuation

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***Abstract*—It is interesting and challenging to investigate the problem of prescribed performance constraints of uncertain nonlinear systems subject to inelastic control input saturation due to their potential conflicts. By designing a novel reference compensation mechanism that depends simultaneously on the input saturation error and the relative distance between performance boundaries and tracking errors, the reference signals are adaptively modified only when absolutely necessary, making the tracking control with prespecified performance attributes affordable even in the presence of strict input constraints. In addition, a robust proportional–integral control scheme featuring with structure simplification and disturbance rejection is developed for a class of nonlinear systems with mismatched uncertainties. It is proved that all the closed-loop signals are bounded, and the proposed method can also ensure the output tracking performance. Numerical simulation results verify and clarify the effectiveness and superiority of the theoretical results.**

***Index Terms*—Input saturation, low complexity, nonlinear systems, tracking performance.**

## I. Introduction

Practical systems usually suffer from different constraints originating from physical limitations and task allocations, including output/performance constraints and input saturation [1], [2], [3]. The controlled systems may be damaged, or even a serious accident may occur once such constraints are violated. Thus, constrained control of complex systems has aroused much interest in control communities that address the challenges of designing controllers for systems with inherent nonlinearities and external uncertainties while adhering to various physical or operational constraints. A series of control techniques, such as prescribed performance control [4], barrier functions [5], and funnel control [6], has been developed to handle these constraints, often leveraging optimization and Lyapunov-based methods to guarantee closed-loop stability and constraint satisfaction.

Performance constraints mean that the evolution of output tracking error is confined within some quantified boundaries characterizing an arbitrarily fast decaying rate and an arbitrarily small steady-state error. It was systematically solved adaptively and robustly by the pioneering works [4] and [7], respectively. Subsequently, prescribed performance control of nonlinear systems has gained much attention, stemming from single system [8], [9], [10] to networked systems [11], [12]. In [8], an adaptive performance guaranteed control scheme was designed for parameterized nonlinear systems with actuator failures. In the same spirit, a proportional–integral (PI) robust adaptive control strategy was developed to achieve desired tracking performance without the trial-and-error procedure [9], but the considered systems were restricted to the Brunovsky canonical form rather than the more general strict-feedback one. To ease this limitation, Zhang and Chai [13] proposed a PI control approach for lower triangular nonlinear systems with prespecified performance attributes. It is worth noting that the preferred tracking performance is achieved by the aforementioned works provided any control efforts. However, the control inputs of the real-world systems are typically constrained due to physical conditions, which may cause the foregoing performance guaranteed control schemes to fail in the presence of input saturation.

It is well known that the performance of controlled systems subject to inelastic input constraints would be significantly degraded and even be violated, leading to internal instability. To deal with this issue, a funnel controller was developed for normal form nonlinear systems with input saturation and arbitrary relative degree [14]. A flexible performance-based control scheme was devised for nonlinear systems by modifying the performance boundaries when the control input was saturated [15]. Following the same vein, the finite-time tracking performance was ensured for the input-constrained nonlinear systems with unknown control coefficients [16]. Notably, the nonlinear functions in [15] and [16] are assumed to be known, and the computational complexity is burdensome. Taking uncertainties into consideration, a switching control approach was devised in [17] to enforce the tracking error to evolve strictly inside a relaxed performance envelope as the input saturation took place. In addition, a low complexity control law was proposed for uncertain nonlinear systems with input constraints [18], which enabled the closed-loop system to be stabilized via reference modification. Specifically, the reference trajectories are temporarily modified whenever the control input is saturated to make the tracking for the modified references feasible and further exclude the internal instability. There is, however, an issue that naturally arises with the poor tracking performance due to the frequent modification of reference trajectories.

Enlightened by such observations, it is obvious that prescribed performance control of input-constrained uncertain nonlinear systems still deserves further investigation. To this end, a robust PI performance guaranteed control scheme is developed in this work. The main contributions are extracted as follows.

1) Resorting to the relative distance mechanism, a novel reference compensation strategy is designed in the sense that the reference signals are adaptively modified whenever it is absolutely necessary, instead only of the appearance of control input saturation, making

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the proposed compensation strategy more user-friendly and efficient compared with [15], [16], and [18].

2) The steady-state oscillation phenomenon of the tracking errors is effectively suppressed with the proposed PI control scheme. Moreover, in the presence of inelastic input constraints, the prespecified performance is ensured for nonlinear systems with mismatched uncertainties.
3) The PI control scheme does not involve any prior knowledge of system uncertainties, and no approximation operators, such as fuzzy logic systems [16], [19], neural networks, and adaptive laws [20], are used, leading to a low-complexity solution.

The standard notations are utilized in this work, with $\mathbb{R}$ for the set of real numbers, $\mathbb{R}^n$ for the set of $n$-dimensional real vectors, $|*|$ for the absolute value of a real number $*$, $\mathcal{I}_j^k$ with $j<k$ for the integer index set $\{j,\ldots,k\}$, and $\mathrm{int}(\bullet)$ and $\partial(\bullet)$ for the interior and boundary of a compact set $\bullet$, respectively. The remainder of this work is outlined below. We formulate the problem in Section II. We present the control design in Section III, closely followed by its stability analysis in Section IV. We provide simulation verification in Section V. Finally, we conclude in Section VI.

## II. Problem Formulation

### A. System Description

Consider the following uncertain strict-feedback nonlinear systems with input saturation:

$$\begin{cases}\dot{x}_i(t)=g_i\left(\bar{x}_i(t),d_i(t)\right)x_{i+1}(t)+f_i\left(\bar{x}_i(t),d_i(t)\right), i\in\mathcal{I}_1^{n-1}\\ \dot{x}_n(t)=g_n\left(\bar{x}_n(t),d_n(t)\right)\mathrm{sat}(u(t))+f_n\left(\bar{x}_n(t),d_n(t)\right)\\ y(t)=x_1(t)\end{cases}\tag{1}$$

where $\bar{x}_i=[x_1,\ldots,x_i]^{\mathrm{T}}\in\mathbb{R}^i$, $i\in\mathcal{I}_1^n$; $\bar{x}_n\in\mathbb{R}^n$ is the system state vector; $d_i\in\mathbb{R}^m$ denote the bounded and piecewise continuous disturbances; $u\in\mathbb{R}$ and $y\in\mathbb{R}$ are the control input and system output, respectively; $g_i(\cdot)\in\mathbb{R}$ and $f_i(\cdot)\in\mathbb{R}$ are locally Lipschitz on $\bar{x}_i$ and piecewise continuous on $d_i$ with totally unknown expressions;[1] $\mathrm{sat}(u)\in\mathbb{R}$ is the saturation function defined by Gao et al. [21]

$$\mathrm{sat}(u)=\begin{cases}\mathrm{u}, & |u|<u_{\max}\\ \mathrm{sign}(u)u_{\max}, & |u|\geq u_{\max}\end{cases}\tag{2}$$

where $u_{\max}>0$ represents the saturation threshold.

Some common assumptions are made as follows.

*Assumption 1:* There exists an unknown constant $\underline{g}$ such that $|g_i(\cdot)|\geq\underline{g}>0$, $i\in\mathcal{I}_1^n$. Without loss of generality, we consider that $g_i(\cdot)\geq\underline{g}>0$.

*Assumption 2:* The desired trajectory $y_r(t)\in\mathbb{R}$ is continuously differentiable and uniformly bounded so that $|y_r(t)|\leq\bar{y}_r\ \forall t\geq 0$, with $\bar{y}_r$ being a positive constant. Furthermore, its first-order derivative $\dot{y}_r(t)$ is also bounded, although unknown.

*Assumption 3 (See [22]):* For a practical plant modeled by system (1) subject to input saturation (2), there exists a feasible control input $u$ to solve the tracking control problem stated in Section II-B.

*Remark 1:* Assumption 1 is called the global controllability condition [15], [18]. Assumption 2 indicates that only the boundedness of the reference trajectory is needed, and its derivatives will not be utilized in the control design. Assumption 3 is reasonable, since an unstable system cannot be globally stabilized when the input saturation occurs [23].

[1]In this article, a function $F(a,b)$ may be abbreviated as $F(\cdot)$ or $F$ when the arguments are clear from the context.

### B. Problem Statement

To exclude the instability caused by the violation of the performance funnel within which the tracking error evolves in the company of control input saturation, a potential solution is to compensate the reference trajectory, enabling the tracking objective to be affordable. Therefore, the problem dealt with in this work reads as follows.

*Control problem:* Consider system (1) with the input constraint (2) and Assumptions 1–3. Design a compensated reference based robust PI control scheme such that: 1) all signals in the closed-loop are bounded; 2) the tracking error $\tilde{e}(t):=y(t)-\tilde{y}_r(t)$ evolves strictly inside the set

$$\mathcal{F}:=\{\tilde{e}(t)\in\mathbb{R}\,||\tilde{e}(t)|<\rho_1(t)\}$$

where $\tilde{y}_r(t)$ is the compensated reference signal constructed via (4), (5), and $\rho_1(t)=(\rho_1^0-\rho_1^\infty)e^{-\kappa_1 t}+\rho_1^\infty$ denotes the output performance function with $\rho_1^0>\rho_1^\infty>0$, and $\kappa_1>0$.

*Remark 2:* It should also be pointed out that the output performance attributes of system (1) are solely determined by the design parameters $\kappa_1$ and $\rho_1^\infty$ embedded in the performance function $\rho_1(t)$. Specifically, the minimum convergence rate and the maximum steady-state error are quantified by $\kappa_1$ and $\rho_1^\infty$, respectively [7]. A larger value of $\kappa_1$ leads to a faster convergence rate of the tracking error $\tilde{e}(t)$, and a smaller $\rho_1^\infty$ results in better steady-state tracking behavior. Therefore, forcing the output tracking error to evolve strictly inside $\mathcal{F}$ introduces specific transient and steady-state performance bounds.

## III. Control Design

### A. Reference Compensation Mechanism Design

To compensate for the reference trajectory in the presence of input saturation, we first define the saturation error $\breve{u}(t)$ as

$$\breve{u}(t):=\mathrm{sat}(u(t))-u(t).\tag{3}$$

Motivated by [18], the increments $\delta_i(t)$ $(i\in\mathcal{I}_1^n)$ are generated by

$$\dot{\delta}_i(t)=-\iota_i\delta_i(t)+\delta_{i+1}(t),\quad \delta_i(0)=0,\quad i\in\mathcal{I}_1^{n-1}\tag{4a}$$

$$\dot{\delta}_n(t)=-\iota_n\delta_n(t)+\breve{u}(t)\omega,\quad \delta_n(0)=0\tag{4b}$$

$$\omega(t)=(\mathrm{sign}(\psi(t)-\lambda)-1)(\psi-\lambda)\tag{4c}$$

$$\psi(t)=\rho_n(t)-|x_n(t)-\alpha_{n-1}(t)|\tag{4d}$$

where $\iota_i$ and $\lambda$ are positive design parameters and $\rho_n(t)$ and $\alpha_{n-1}(t)$ denote the last performance function and the last virtual control input, respectively, which will be given in the sequel. Thus, as part of the control design, the compensated reference $\tilde{y}_r(t)$ for the output $y(t)$ to track is specified as

$$\tilde{y}_r(t)=y_r(t)+\delta_1(t).\tag{5}$$

*Remark 3:* Note that the relative distance between the performance function $\rho_n(t)$ and the corresponding tracking error are described by the variable $\psi(t)$, while the relative safe distance is dominated by the threshold parameter $\lambda$. It is elaborated from (3) and (4) that if the control input is not saturated ($\breve{u}(t)\equiv 0$), or the relative safe distance threshold is never triggered ($\psi(t)\geq\lambda\,\forall t\geq 0$), the increment signals $\delta_i(t)$ $(i\in\mathcal{I}_1^n)$ remain zero. In other words, the compensation mechanism of reference trajectory is activated only if the input is saturated and the relative distance is smaller than the designed safe threshold $\lambda$, making the reference compensation absolutely necessary. This excludes the phenomenon that the actual tracking performance has to be degraded even if such saturation does not lead to internal instability of the considered system. Practically, the prescribed performance may also be ensured in the case that the input is not severely saturated, which reveals the strong conservatism of the compensation methods based solely on input saturation error [15], [18]. However, the elaborately proposed compensation mechanism relaxes it by embedding additional relative distance thresholds (4c) and (4d), essentially distinguishing itself from the methods developed in [15] and [18].

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*Remark 4:* It is derived from (4c) that $\omega(t)$ is upper bounded by $2\lambda$, thus the amplitude of the increment signals $\delta_i(t)$ ($i \in \mathcal{I}_1^n$) mainly depends on that of $2\lambda\breve{u}(t)$ if $\iota_i$ is specified, i.e., larger $2\lambda|\breve{u}(t)|$ leads to larger $|\delta_i(t)|$. Likewise, if the compensation mechanism is temporarily activated, then $\delta_i(t)$ will converge to zero at the rate of $e^{-\iota_i t}$ once such a mechanism is ceased. Therefore, the amplitude and convergence rate of $\delta_i(t)$ are adjustable via design parameters $\lambda$ and $\iota_i$, respectively.

### B. Robust PI Control Design

To ensure prescribed performance control, we define a continuously differentiable barrier function $\mathcal{T}: (-1, 1) \to \mathbb{R}$ with $\lim_{\star \to -1^+} \mathcal{T}(\star) = -\infty$ and $\lim_{\star \to 1^-} \mathcal{T}(\star) = +\infty$. The immediate candidates include $\mathcal{T}(\star) = \ln\left(\frac{1+\star}{1-\star}\right)$ and $\mathcal{T}(\star) = \tan(\frac{\pi}{2}\star)$. The latter is used in this work.

We now design the recursive robust PI control scheme as follows:

$$z_i(t) = \frac{x_i(t) - \alpha_{i-1}(t) - \delta_i(t)}{\rho_i(t)}, \quad i \in \mathcal{I}_1^n \tag{6}$$

$$\epsilon_i(t) = \mathcal{T}(z_i(t)), \quad i \in \mathcal{I}_1^n \tag{7}$$

$$\alpha_i(t) = -k_{i1}\epsilon_i(t) - k_{i2}\int_0^t \epsilon_i(\nu)d\nu, \quad i \in \mathcal{I}_1^{n-1} \tag{8}$$

$$u(t) = -k_{n1}\epsilon_n(t) \tag{9}$$

where $\alpha_0(t) = y_r(t)$, $k_{i1}$ and $k_{i2}$ are positive constants and $\rho_i(t) = (\rho_i^0 - \rho_i^\infty)e^{-\kappa_i t} + \rho_i^\infty$ are performance functions with $\rho_i^0 > |x_i(0) - \alpha_{i-1}(0)|$, $\rho_i^0 > \rho_i^\infty > 0$, and $\kappa_i > 0$. Note that the incremental signal $\delta_i(t)$ ($i \in \mathcal{I}_1^n$) are introduced into the normalized errors $z_i(t)$ defined by (6), which provides a solid foundation for achieving prespecified performance control of uncertain nonlinear systems subject to inelastic input saturation.

*Remark 5:* Regarding the robust control law shown above, some interesting features are discussed below. First, it preserves the simplicity in the structure of the traditional PI control scheme [9], [13]. Second, there is neither any prior core knowledge of system uncertainties nor approximation operators, such as neural networks and fuzzy logic systems, used to obtain such knowledge, making it different from [16]. Third, the time derivatives of reference trajectory $y_r(t)$ and intermediate control input $\alpha_i(t)$ ($i \in \mathcal{I}_1^{n-1}$) are not incorporated into the control scheme, although it is designed in a recursive manner, distinguishing itself from [20]. Overall, such threefold efforts result in a low complexity control design.

*Remark 6:* Since the performance functions $\rho_i(t)$ ($i \in \mathcal{I}_2^n$) are not directly related to the output tracking performance, they can be freely selected, provided $\rho_i^0 > |x_i(0) - \alpha_{i-1}(0)|$. However, a careful inspection of the simulation results reveals that their selections have a nonnegligible impact on the evolution of the output tracking error inside $\mathcal{F}$ as well as the amplitude of the control input. In implementation, there should thus be a tradeoff between the acceptable control effort and the performance attributes.

## IV. Main Results

The main results of this article are summarized into the following theorem.

*Theorem 1:* Consider system (1) satisfying Assumptions 1–3. Let $\kappa_1 > 0$, $\rho_1^\infty > 0$ denote the user-defined minimum convergence rate and the maximum steady-state tracking error, respectively. For any $\varpi \geq 0$, the controller designed via (3)–(9) with

$$u_{\max} \geq \max\left\{|u(0)| - \varpi, \sup_{t\geq 0}\frac{|f_n(\cdot) - \dot{\alpha}_{n-1} - z_n\dot{\rho}_n|}{\underline{g}}\right\}$$

*guarantees:*

a) the boundedness of all signals in the closed-loop;

b) $\tilde{e}(t) \in \mathcal{F}\,\forall t \geq 0$.

*Proof:* Let $\bar{z}_i = [z_1, \ldots, z_i]^{\mathrm{T}} \in \mathbb{R}^i, \bar{\delta}_i = [\delta_1, \ldots, \delta_i]^{\mathrm{T}} \in \mathbb{R}^i$. From (1), (2), (4), and (6)–(8), we have

$$\begin{cases} \dot{z}_i(t) := h_i(\bar{z}_{i+1}, \bar{\delta}_{i+1}, t) = \frac{1}{\rho_i}\left(g_i(\cdot)x_{i+1} + f_i(\cdot)\right. \\ \quad \left. - \dot{\alpha}_{i-1} + \iota_i\delta_i - \delta_{i+1} - z_i\dot{\rho}_i\right), \quad i \in \mathcal{I}_1^{n-1} \\ \dot{z}_n(t) := h_n(\bar{z}_n, \bar{\delta}_n, t) = \frac{1}{\rho_n}\left(g_n(\cdot)\mathrm{sat}(u) + f_n(\cdot)\right. \\ \quad \left. - \dot{\alpha}_{n-1} + \iota_n\delta_n - \breve{u}\omega - z_n\dot{\rho}_n\right) \end{cases} \tag{10}$$

with

$$\begin{cases} x_1(t) = z_1(t)\rho_1(t) + \delta_1(t) + y_r(t) \\ x_i(t) = z_i(t)\rho_i(t) + \delta_i(t) - k_{(i-1)1}\mathcal{T}(z_{i-1}(t)) \\ \quad -k_{(i-1)2}\int_0^t \mathcal{T}(z_{i-1}(\nu))d\nu, \quad i \in \mathcal{I}_2^n. \end{cases} \tag{11}$$

By recalling the definition of $\bar{z}_i$ and $\bar{\delta}_i$, the dynamics given in (4) and (10) can be compactly rewritten as

$$\dot{\bar{z}}_n(t) := H_1(\bar{z}_n, \bar{\delta}_n, t) = [h_1, \ldots, h_n]^{\mathrm{T}}$$

$$\dot{\bar{\delta}}_n(t) := H_2(\bar{z}_n, \bar{\delta}_n, t) = \left[\dot{\delta}_1, \ldots, \dot{\delta}_n\right]^{\mathrm{T}}.$$

Let $\xi = [\bar{z}_n^{\mathrm{T}}, \bar{\delta}_n^{\mathrm{T}}]^{\mathrm{T}}$. It then follows that:

$$\dot{\xi}(t) := H(\xi, t) = \left[H_1^{\mathrm{T}}(\xi, t), H_2^{\mathrm{T}}(\xi, t)\right]^{\mathrm{T}}. \tag{12}$$

Define the open set

$$\Omega_\xi = (-1, 1)^n \times \mathbb{R}^n.$$

The $\xi$-dynamics in (12) is well-defined for $\xi(t) \in \Omega_\xi$. The remainder of the proof is divided into the following three phases.

*Phase 1—Existence and uniqueness of a maximal solution $\xi(t)$:* $[0, \tau_{\max}) \to \Omega_\xi$*:* Since $\delta_i(0) = 0$ and $\rho_i^0 > |x_i(0) - \alpha_{i-1}(0)|, i \in \mathcal{I}_1^n$, one has $|z_i(0)| < 1$. There thus exists a constant $\varpi \geq 0$ such that

$$|u(0)| \leq u_{\max} + \varpi \tag{13}$$

which further indicates that

$$|\epsilon_n(0)| \leq \frac{u_{\max} + \varpi}{k_{n1}} < \infty$$

$$|z_n(0)| \leq \mathcal{T}^{-1}\left(\frac{u_{\max} + \varpi}{k_{n1}}\right) < 1. \tag{14}$$

Hence, $\xi(0) \in \Omega_\xi$. In addition, owing to the boundedness and piecewise continuity of $y_r(t)$, $\dot{y}_r(t)$, $\rho_i(t)$, and $\dot{\rho}_i(t)$, to locally Lipschitz continuity of $g_i(\cdot)$ and $f_i(\cdot)$, and to the differentiability of $\alpha_i(t)$ over the set $\Omega_\xi$, it is straightforward to get from (4) and (10) that the dynamics given in (12) satisfies the hypotheses of [24, Thm. 1]. Therefore, the existence and uniqueness of a maximal solution $\xi(t) : [0, \tau_{\max}) \to \Omega_\xi$ are ensured with $0 < \tau_{\max} < \infty$.

*Phase 2—Boundedness of all signals in the closed-loop:* Define the set

$$\Omega_\epsilon = \{\epsilon_n(t) \in \mathbb{R}\,|k_{n1}\epsilon_n(t)| \leq u_{\max} + \varpi\}. \tag{15}$$

As $\epsilon_n(0) \in \Omega_\epsilon$, there exists a time interval $\breve{\tau} \subseteq [0, \tau_{\max})$ with $0 \in \breve{\tau}$ such that

$$\epsilon_n(t) \in \Omega_\epsilon \quad \forall t \in \breve{\tau}.$$

It in turn implies from (3) that

$$|\breve{u}(t)| \leq \varpi \quad \forall t \in \breve{\tau}.$$

According to (4), we have

$$|\delta_i(t)| \leq \frac{\varpi\omega(t)}{\Pi_{j=i}^n \iota_j} \quad \forall t \in \breve{\tau} \tag{16}$$

For ease of description, let

$$\beta_i(t) = \cos^2\left(\frac{\pi z_i(t)}{2}\right), \quad p_i(t) = \frac{\pi}{2\beta_i(t)\rho_i(t)}, \quad i \in \mathcal{I}_1^n.$$

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It is evident that there exist constants $\underline{p}$ and $\overline{p}$ such that

$$0 < \underline{p} \le p_i(t) \le \overline{p} < \infty \quad \forall \xi(t) \in \Omega_\xi.$$

To proceed, the proof follows a recursive procedure.

*Step 1: ($t \in \check{\tau}$):* Taking the time derivative of $\epsilon_1(t)$ given in (7) upon using (10) yields that

$$\dot{\epsilon}_1(t) = p_1(t)\left(g_1(\cdot)\alpha_1(t) + q_1(t)\right) \tag{17}$$

where

$$q_1 = g_1(\cdot)(z_2\rho_2 + \delta_2) + f_1(\cdot) - \dot{y}_r + \iota_1\delta_1 - \delta_2 - z_1\dot{\rho}_1.$$

Define a Lyapunov function candidate as

$$V_1(t) = \frac{1}{2}\epsilon_1^2(t).$$

By (8) and (17), the time derivative of $V_1$ becomes

$$\dot{V}_1(t) = p_1(t)\epsilon_1(t)\left(-k_{11}g_1(\cdot)\epsilon_1(t) + r_1(t) + q_1(t)\right) \tag{18}$$

with

$$r_1(t) = -k_{12}g_1(\cdot)\int_0^t \epsilon_1(\nu)d\nu.$$

On one hand, since $|z_i(t)| < 1\ \forall t \in \check{\tau}, i \in \mathcal{I}_1^n$, it follows that $r_1$ is bounded for $t \in \check{\tau}$. On the other hand, the boundedness of $y_r$, $\dot{y}_r$, $\rho_1$, $\dot{\rho}_1$, $\rho_2$, $\delta_1$, and $\delta_2$ is ensured by assumption and construction, we know from (11) that $x_1$ is bounded for $t \in \check{\tau}$, so is $q_1$ given in (17) owing to the continuity of $g_1(\cdot)$ and $f_1(\cdot)$. Denote

$$\bar{q}_1 := \sup_{t\in\check{\tau}}|q_1(t)|, \quad \bar{r}_1 := \sup_{t\in\check{\tau}}|r_1(t)|.$$

It is readily derived from Assumption 1 and (18) that

$$\dot{V}_1(t) \le p_1(t)\left(-k_{11}\underline{g}|\epsilon_1(t)|^2 + (\bar{r}_1 + \bar{q}_1)|\epsilon_1(t)|\right)$$

which indicates that $\dot{V}_1(t) \le 0$ if $|\epsilon_1(t)| \ge \frac{\bar{r}_1+\bar{q}_1}{k_{11}\underline{g}}$. Therefore

$$|\epsilon_1(t)| \le \bar{\epsilon}_1 := \max\left\{|\epsilon_1(0)|, \frac{\bar{r}_1 + \bar{q}_1}{k_{11}\underline{g}}\right\} \quad \forall t \in \check{\tau}. \tag{19}$$

It is thus readily derived that

$$|z_1(t)| \le \mathcal{T}^{-1}(\bar{\epsilon}_1(t)) < 1 \quad \forall t \in \check{\tau}.$$

Accordingly, the boundedness of $\alpha_1(t)$ and its time derivative $\dot{\alpha}_1(t) = -k_{11}\dot{\epsilon}_1(t) - k_{12}\epsilon_1(t)$ can be ensured in light of (17) and (19).

Note from (18) that $\dot{V}_1(t)$ without the integral term ($k_{12} = 0$) can be calculated as

$$\begin{aligned}\dot{V}_1^{\mathrm{P}}(t) &= -k_{11}g_1(\cdot)p_1\epsilon_1^2(t) + p_1(t)q_1(t)\epsilon_1(t)\\ &\le -k_{11}\underline{g}\underline{p}\epsilon_1^2(t) + \overline{p}\bar{q}_1|\epsilon_1(t)|\\ &\le -\frac{k_{11}\underline{g}\underline{p}}{2}\epsilon_1^2(t) + \frac{\overline{p}\bar{q}_1^2}{2k_{11}\underline{g}}\\ &\le -c_1V_1^{\mathrm{P}}(t) + d_1\end{aligned} \tag{20}$$

where $c_1 = k_{11}\underline{g}\underline{p}$ and $d_1 = \frac{\overline{p}\bar{q}_1^2}{2k_{11}\underline{g}}$. However, $\dot{V}_1(t)$ with the integral term ($k_{12} \ne 0$) is

$$\begin{aligned}\dot{V}_1^{\mathrm{PI}}(t) &= \dot{V}_1(t)\\ &= \dot{V}_1^{\mathrm{P}}(t) - k_{12}p_1g_1(\cdot)\epsilon_1(t)\int_0^t \epsilon_1(\nu)d\nu\ \forall t \in \check{\tau}.\end{aligned} \tag{21}$$

Since it is unknown whether $\epsilon_1(t)\int_0^t \epsilon_1(\nu)d\nu$ is always positive for all $t \in \check{\tau}$, we define $t_0 = 0$ and $t_k$ ($t_k < t_{k+1}, k = 1, 2, \ldots$) such that $\epsilon_1(t_k) = 0$. It then follows that:

$$k_{12}p_1(t)g_1(\cdot)\epsilon_1(t)\int_{t_k}^t \epsilon_1(\nu)d\nu \ge 0 \quad \forall t \in [t_k, t_{k+1}). \tag{22}$$

The equality holds if and only if $t = t_k$. For $t \in (t_k, t_{k+1})$, it holds that $k_{12}p_1(t)g_1(\cdot)\epsilon_1(t)\int_{t_k}^t \epsilon_1(\nu)d\nu > 0$. Therefore, it is derived from (21) that

$$\dot{V}_1^{\mathrm{PI}}(t) < \dot{V}_1^{\mathrm{P}}(t) \quad \forall t \in (t_k, t_{k+1}) \tag{23}$$

which implies that the attraction region of $V_1^{\mathrm{PI}}(t)$ is smaller than that of $V_1^{\mathrm{P}}(t)$ over $(t_k, t_{k+1})$. Accordingly, $\epsilon_1(t)$ with the proposed PI controller will evolve strictly inside a smaller compact set than that with the traditional proportional feedback over $(t_k, t_{k+1})$. As a result, reduced errors $z_1(t)$ and $\tilde{e}(t)$ are obtained over $(t_k, t_{k+1})$.

*Step i. ($i \in \mathcal{I}_2^{n-1}$, $t \in \check{\tau}$):* Combining (7) and (10) yields that

$$\dot{\epsilon}_i(t) = p_i(t)\left(g_i(\cdot)\alpha_i(t) + q_i(t)\right)$$

with

$$\begin{aligned}q_i = {} & g_i(\cdot)(z_{i+1}\rho_{i+1} + \delta_{i+1}) + f_i(\cdot) - \dot{\alpha}_{i-1}\\ & + \iota_i\delta_i - \delta_{i+1} - z_i\dot{\rho}_i.\end{aligned}$$

Similar to (18) and (19) shown in Step 1, by defining $V_i(t) = \frac{1}{2}\epsilon_i^2(t)$, we have

$$|\epsilon_i(t)| \le \bar{\epsilon}_i := \max\left\{|\epsilon_i(0)|, \frac{\bar{r}_i + \bar{q}_i}{k_{i1}\underline{g}}\right\} \quad \forall t \in \check{\tau}$$

which implies that

$$|z_i(t)| \le \mathcal{T}^{-1}(\bar{\epsilon}_i(t)) < 1 \quad \forall t \in \check{\tau}.$$

Thus, the boundedness of $\alpha_i$ and $\dot{\alpha}_i$ is ensured. Along with (20)–(23), smaller errors $\epsilon_i(t)$ and $z_i(t)$ are derived with the proposed PI control than that only with the proportional one.

*Step n. ($t \in \check{\tau}$):* According to (11), it is easy to conclude that $x_n$ is bounded. To proceed, the following two cases should be considered.

*Case 1— $|u(t)| < u_{\max}$:* Based on (9), (14), and (15), one has $\epsilon_n(t) \in \mathrm{int}(\Omega_\epsilon)$, and

$$|z_n(t)| < \mathcal{T}^{-1}\left(\frac{u_{\max}}{k_{n1}}\right) \le \mathcal{T}^{-1}\left(\frac{u_{\max}+\varpi}{k_{n1}}\right) < 1.$$

*Case 2— $|u(t)| \ge u_{\max}$:* Form (7) and (10), the time derivative of $\epsilon_n(t)$ is computed as

$$\begin{aligned}\dot{\epsilon}_n &= p_n\left(g_n(\cdot)\mathrm{sat}(u) + \iota_n\delta_n - \breve{u}\omega + q_n\right)\\ &= p_n\left(g_n(\cdot)u_{\max}\mathrm{sign}(u) + \iota_n\delta_n - \breve{u}\omega + q_n\right)\end{aligned}$$

where $q_n = f_n(\cdot) - \dot{\alpha}_{n-1} - z_n\dot{\rho}_n$ is bounded. Note from (9) that

$$\mathrm{sign}(u) = -\mathrm{sign}(\epsilon_n). \tag{24}$$

Select a Lyapunov function candidate as

$$V_n(t) = \frac{1}{2}\epsilon_n^2(t)$$

then the time derivation of $V_n$ becomes

$$\dot{V}_n = p_n\epsilon_n\left(-g_n(\cdot)u_{\max}\mathrm{sign}(\epsilon_n) + \iota_n\delta_n - \breve{u}\omega + q_n\right).$$

If $\epsilon_n(t) \in \partial(\Omega_\epsilon)$, then it follows from (15) that $|u(t)| = u_{\max} + \varpi$ and $|\breve{u}(t)| = \varpi$. As $\breve{u} = \mathrm{sat}(u) - u$ and $|\mathrm{sat}(u)| \le |u|$, there holds that $\mathrm{sign}(\breve{u}) = -\mathrm{sign}(u)$ if $\varpi \ne 0$, and $\breve{u} = 0$ otherwise. Thus, from (24), we have $\breve{u} = \varpi\mathrm{sign}(\epsilon_n)$, which further yields that

$$\begin{aligned}\dot{V}_n &\le p_n|\epsilon_n|\left(-\underline{g}u_{\max} + \iota_n|\delta_n| - \varpi\omega + |q_n|\right)\\ &\le p_n|\epsilon_n|\left(-\underline{g}u_{\max} + |q_n|\right)\end{aligned} \tag{25}$$

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where $\iota_n|\delta_n| - \varpi\omega \leq 0$ by (16) has been used. Also, the introduction of $\omega$ does not affect the system internal stability. Therefore, if

$$u_{\max} \geq \sup_{t\in\check{\tau}} \frac{|q_n(t)|}{\underline{g}} = \sup_{t\in\check{\tau}} \frac{|f_n(\cdot) - \dot{\alpha}_{n-1} - z_n\dot{\rho}_n|}{\underline{g}} \tag{26}$$

we have $\dot{V}_n \leq 0$ as $\epsilon_n \in \partial(\Omega_\epsilon)$. It thus can be concluded by combining Cases 1 and 2 that the set $\Omega_\epsilon$ is positive invariant if (26) holds. Owing to $\epsilon_n(0) \in \Omega_\epsilon$, it follows that $\check{\tau} \equiv [0, \tau_{\max})$, and there holds that

$$\epsilon_n(t) \in \Omega_\epsilon \quad \forall t \in [0, \tau_{\max})$$

which in turn indicates that

$$|z_n(t)| \leq \mathcal{T}^{-1}\left(\frac{u_{\max} + \varpi}{k_{n1}}\right) < 1 \quad \forall t \in [0, \tau_{\max}).$$

As a result, the boundedness of $u(t)$ over $[0, \tau_{\max})$ is ensured. Therefore, all signals in the closed-loop system are bounded over $[0, \tau_{\max})$.

*Phase 3—Extension of $\tau_{\max} = \infty$:* The foregoing analysis infers that there exist constants $\bar{z}$ and $\underline{z}$ such that

$$z_i(t) \in \Omega_z = \{z_i(t) | -1 < \underline{z} \leq z_i(t) \leq \bar{z} < 1\}$$

for $t \in [0, \tau_{\max}), i \in \mathcal{I}_1^n$, where $\Omega_z$ is a nonempty and compact subset of $\Omega_\xi$. Assuming $\tau_{\max} < \infty$, it follows from [24, Thm. 2] that there exists a time instant $t' \in [0, \tau_{\max})$ such that $z_i(t') \notin \Omega_z$, which is a contradiction. Thus, $\tau_{\max} = \infty$. Therefore, all closed-loop signals are bounded over $[0, \infty)$.

What remains to be shown is the achievement of prescribed output performance. It follows from (5) and (11) that $|y(t) - \tilde{y}_r(t)| < \rho_1(t)\ \forall t \in [0, \infty)$, which directly indicates that, in the presence of inelastic control input saturation (2), the compensated tracking error $\tilde{e}(t)$ is strictly confined within the performance funnel $\mathcal{F} := \{\tilde{e}(t) \in \mathbb{R} | |\tilde{e}(t)| < \rho_1(t)\}$ if the conditions (13) and (26) hold. The proof is complete. ■

*Remark 7:* The feasible condition in (26) shows the reasonable dependence relationship between the lower bounds of saturation level $u_{\max}$ and uncertain control gains, external disturbances/uncertainties, and the performance attributes. For a practical plant, it is natural that a larger saturation amplitude is required for smaller control gains, larger disturbances/uncertainties, or smaller steady-state tracking error. Note that the feasible condition is sufficient but not necessary due to the involvement of the supremum operators in (26).

## V. Simulation

To illustrate the effectiveness of the proposed control scheme, we in this section conduct a numerical simulation on the following system:

$$\begin{cases} \dot{x}_1(t) = x_2(t) - 1.5x_1^2(t) - 0.5x_1^3(t) \\ \dot{x}_2(t) = \mathrm{sat}(u(t)) + d(t) \\ y(t) = x_1(t) \end{cases} \tag{27}$$

where $\bar{x}_2 = [x_1, x_2]^{\mathrm{T}}$ and $u$ are the system state and desired control input, respectively. It is noted that system (27) is originated from the jet engine compressor with no-stall model [25]. Since we suppose in Assumption 1 that $g_i > 0$, the control coefficient of the first subsystem is specified as 1. Moreover, the second subsystem is encountered by the nonvanishing random sine disturbance $d(t) = 4\sin(2t)$ in the presence of saturated input $\mathrm{sat}(u)$.

In implementation, the reference trajectory is set as $y_r(t) = \sin(t)$, the performance functions are selected as $\rho_1(t) = 3e^{-t} + 0.01$, and $\rho_2(t) = 4e^{-t} + 0.01$, the saturation level is constrained by $u_{\max} = 3$, the design parameters are chosen as $\iota_1 = 2$, $\iota_2 = 4$, $k_{11} = k_{21} = 2$, $k_{12} = 1$, and $\lambda = 0.1$. In addition, the initial condition is specified as $\bar{x}_2(0) = [-0.8, 0.2]^{\mathrm{T}}$. In order to verify the superiority of the proposed method, we consider the control scheme in [18] for comparison. Thus, we set $\omega = 1$ and $k_{12} = 0$ with other performance parameters and initial conditions remaining unaltered for fairness to rerun the simulation.

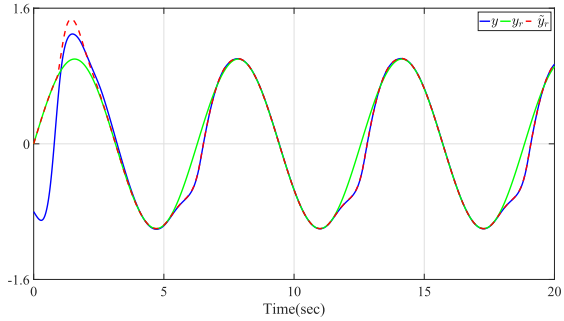



Fig. 1. Evolution of $y$, $y_r$, and $\tilde{y}_r$ under the proposed scheme.

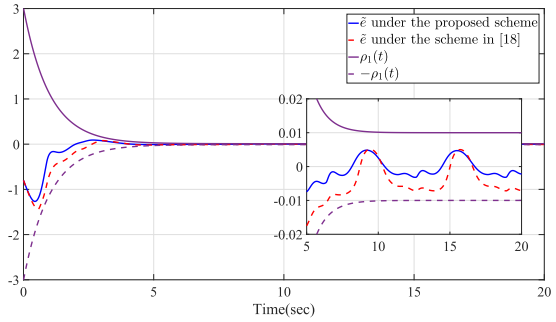



Fig. 2. Evolution of $\tilde{e}$ under different control schemes.

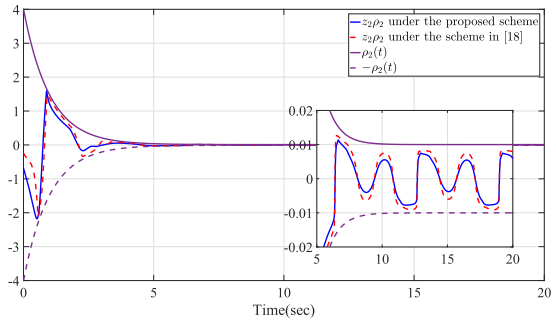



Fig. 3. Evolution of $z_2\rho_2$ under different control schemes.

The simulation results are depicted in Figs. 1–5. Specifically, Fig. 1 shows that with the proposed method the system output $y(t)$ closely tracks the compensated reference $\tilde{y}_r(t)$, which will approach the desired reference $y_r(t)$ if the input saturation is eased. The comparative evolution of the output tracking error $\tilde{e}$ and the virtual tracking error $z_2\rho_2$ is drawn in Figs. 2 and 3, from which it can be seen that although the performance funnels are never violated under different control schemes, the proposed PI control scheme performs better than that in [18]. The evolution of the increment signals $\delta_i$ ($i \in \mathcal{I}_1^2$) and the control input $u$ with its saturation level $\mathrm{sat}(u)$ is depicted in Figs. 4 and 5, respectively. Note from which that the reference compensation mechanism would not be triggered by the proposed method if the input saturation does not lead to internal instability, which confirms the discussion in Remark 3 that the proposed compensation mechanism is activated only when it is absolutely necessary. In contrast, the compensation mechanism in [18] is triggered as long as the control input is saturated. Generally speaking, all the closed-loop signals are ensured to be bounded with the proposed method, and the output tracking performance can also be guaranteed even in the presence of rigorous input saturation.

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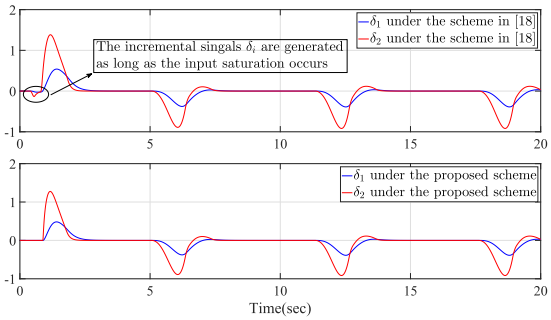



Fig. 4. Evolution of $\delta_1$ and $\delta_2$ under different control schemes.

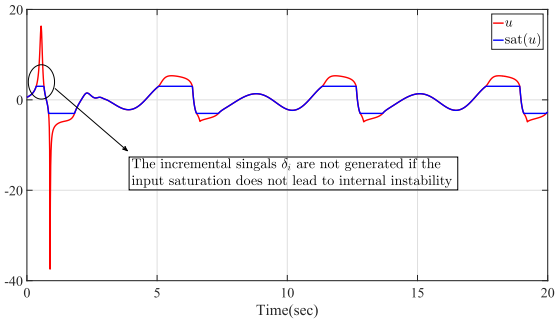



Fig. 5. Evolution of $u$ and $\mathrm{sat}(u)$ under the proposed scheme.

## VI. Conclusion

This article has investigated the problem of performance control of input-constrained nonlinear systems with totally unknown dynamics. By introducing the relative distance mechanism, a novel reference compensation strategy has been skillfully designed such that the reference signals are adaptively modified when it is absolutely necessary. Subsequently, a robust PI control scheme has been proposed featuring low complexity in structure and better output tracking performance in the presence of inelastic input constraints. In the context of extremum seeking [26], how to extend the proposed method to more general nonlinear systems with unknown control directions represents an interesting topic for future study.

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